

Determinants of hypocholesterolemic response to soy and animal protein-based diets.

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The effect of substituting soy for animal protein in mixed diets was determined in young men with mildly elevated plasma cholesterol, 218 to 307 mg/dl. The diets were low in cholesterol, 200 mg/day, with 13 to 16% of energy as protein, 30 to 35% as fat, and a polyunsaturated to saturated fat ratio of 0.5. Of protein 65% was from either mixed animal proteins or isolated soy protein products made comparable by the addition of extracted animal fats. Fresh egg yolk was added to balance the cholesterol content of the diets. Proteins from grains and vegetables were identical in both menus and contributed about 35% of dietary protein. Twenty of 24 subjects decreased plasma cholesterol at the end of the protocol. Subjects were classified as responders or nonresponders as a function of greater or lesser than mean reduction in cholesterol for the groups. Mean decreases in plasma cholesterol, 16 and 13%, for responders in the animal and soy groups were significant, p less than 0.01 and 0.05, respectively. Responders in both groups had higher initial plasma cholesterol values than nonresponders. Although plasma high-density lipoprotein cholesterol decreased slightly, the high-density lipoprotein cholesterol to cholesterol ratio (high-density lipoprotein cholesterol/total cholesterol) remained constant for most individuals. The hypocholesterolemic effects were similar for both animal and soy protein (p less than 0.05) and fat (p less than 0.05) while on the experimental diet. All groups significantly decreased dietary cholesterol (p less than 0.001).